

YTTEROYANE FYR, Norway

WMO: 013040

Lat: 61.572N

Lon: 4.682E

Elev: 25

StdP: 101.03

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.6	-1.3	-10.8	1.5	-0.9	-9.0	1.8	0.3	21.9	4.7	20.4	4.0	5.2	110	0.834

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	20.2	16.3	18.7	15.5	17.5	14.7	17.0	19.6	16.0	18.1	15.1	16.9	3.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.2	18.6	14.9	10.6	17.6	14.0	10.0	16.5	47.9	19.9	45.0	18.3	42.2	17.0	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	17.4	15.7	-3.4	23.0	2.0	3.1	-4.9	25.3	-6.1	27.1	-7.2	28.8	-8.7	31.0		
(5)			-5.3	17.9	2.0	1.4	-6.8	18.9	-7.9	19.7	-9.1	20.6	-10.5	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	3.9	3.4	4.0	6.3	8.8	11.1	13.9	14.9	12.9	9.8	7.1	4.9		
	DBStd	4.56	2.29	2.37	1.96	2.25	2.46	1.99	2.50	2.00	2.01	2.35	2.55	2.76		
	HDD10.0	1023	189	184	187	114	54	10	1	0	2	32	91	159		
	HDD18.3	3620	448	418	445	362	297	218	141	108	163	266	338	418		
	CDD10.0	451	0	0	0	3	16	42	121	153	89	24	3	0		
	CDD18.3	6	0	0	0	0	0	0	3	2	0	0	0	0		
	CDH23.3	7	0	0	0	0	0	0	7	0	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	7.5	9.1	8.6	7.6	6.4	6.3	6.0	5.7	6.3	7.6	8.3	8.8	9.4		
(15) Precipitation	PrecAvg	1948	194	166	155	117	92	104	120	139	189	224	219	228		
	PrecMax	2484	345	308	317	210	156	213	191	280	356	398	374	408		
	PrecMin	1295	28	1	24	22	19	14	20	45	60	98	52	50		
	PrecStd	312	98	82	73	44	37	51	42	58	89	85	100	103		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.4	8.5	15.5	18.2	18.8	23.4	21.8	19.4	16.0	13.1	10.6		
		MCWB	6.3	6.2	5.3	10.9	12.8	13.3	17.8	16.8	14.8	11.7	9.9	8.5		
	2%	DB	8.1	7.5	7.5	12.1	15.2	16.2	20.7	20.0	17.9	14.4	11.7	9.7		
		MCWB	6.5	6.1	5.4	8.0	11.4	12.7	17.0	16.3	14.2	11.1	9.5	8.0		
	5%	DB	7.6	7.0	7.0	10.0	13.3	14.6	18.7	18.8	16.6	13.5	11.0	9.1		
		MCWB	6.2	5.8	5.5	7.1	10.6	12.2	16.0	15.8	13.6	10.9	8.7	7.6		
	10%	DB	7.0	6.5	6.5	8.9	11.8	13.5	17.1	17.7	15.6	12.7	10.2	8.5		
		MCWB	5.7	5.3	5.1	6.5	9.7	11.7	15.0	15.1	13.2	10.4	8.3	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	7.2	7.0	12.0	13.9	14.4	18.5	18.0	16.1	13.2	11.4	9.4		
		MCDB	8.2	7.4	7.3	14.2	17.0	17.2	22.1	20.4	18.4	14.4	12.0	10.0		
	2%	WB	7.3	6.7	6.5	8.7	12.1	13.2	17.2	17.1	15.0	12.4	10.5	8.6		
		MCDB	7.7	7.1	6.9	10.9	14.3	15.3	20.2	19.4	16.8	13.7	11.3	9.4		
	5%	WB	6.6	6.2	5.9	7.7	10.8	12.5	16.1	16.2	14.2	11.6	9.6	8.0		
		MCDB	7.2	6.9	6.6	9.5	12.8	14.1	18.5	18.1	15.9	13.1	10.6	8.9		
	10%	WB	5.9	5.5	5.3	6.9	9.8	11.8	14.8	15.4	13.5	10.7	8.7	7.2		
		MCDB	6.8	6.4	6.2	8.3	11.6	13.3	16.6	17.2	15.2	12.2	9.9	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.5	2.3	2.5	2.8	2.9	2.5	2.7	2.6	2.4	2.4	2.6	2.6		
		MCDBR	2.8	2.7	2.7	4.5	5.1	4.5	5.1	4.0	3.6	2.9	3.0	3.1		
	5% WB	MCWBR	2.8	2.5	2.3	2.4	2.9	2.7	2.8	2.3	2.3	2.4	2.7	3.2		
		MCWBR	2.7	2.7	2.4	3.9	4.7	3.9	5.0	3.4	3.2	2.8	3.0	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.331	0.351	0.374	0.363	0.358	0.372	0.366	0.355	0.331	0.296	0.365		
		taud	2.098	2.236	2.312	2.321	2.424	2.473	2.459	2.472	2.472	2.378	2.167	2.376		
	Ebn at Noon		382	624	757	814	859	870	846	819	757	628	383	229		
		Edh at Noon	39	67	86	103	101	98	97	89	75	57	36	25		
(44) All-Sky Solar Radiation	RadAvg		0.18	0.64	1.66	3.31	4.80	5.09	4.64	3.49	2.02	0.97	0.27	0.11		
	RadStd		0.03	0.12	0.26	0.38	0.35	0.61	0.40	0.40	0.32	0.16	0.06	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	+0.68	N/A	N/A	N/A	N/A	-92	-135	N/A	N/A

Nomenclature: See separate page